

一、质量为 μ 的粒子处于一维谐振子势中，在初始时刻的波函数为

$$\psi(x, t=0) = A(2\varphi_0 + \varphi_1)$$

$$E_n = (n + \frac{1}{2}) \hbar \omega$$

$$\psi_n(x) = N_n e^{-\alpha^2 x^2 / 2} H_n(\alpha x)$$

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725 1~5

(1) 求归一化系数 A 。

(2) 求 $t > 0$ 时刻的波函数 $\psi(x, t)$ 。

(3) 求 $t = T$ 时刻，体系的能量。

(4) 求 x 的平均值。

$$\alpha = \sqrt{\frac{\mu \omega}{\hbar}} \quad N_n = \sqrt{\frac{\alpha}{\sqrt{\pi} 2^n n!}} \quad n = 0, 1, 2, \dots$$

解: (1) $|\psi|^2 = 1$

$$\begin{aligned} \int \psi^* \psi dx &= \int |A|^2 (2\langle \varphi_0 | + \langle \varphi_1 |) (2|\varphi_0\rangle + |\varphi_1\rangle) dx \\ &= \int |A|^2 (4|\varphi_0|^2 + |\varphi_1|^2) dx \\ &= 1 \quad \because \int |\varphi_0|^2 dx = 1 \quad \int |\varphi_1|^2 dx = 1 \end{aligned}$$

$$\therefore 5|A|^2 = 1 \quad A = \frac{1}{\sqrt{5}}$$

(2) $E = (n + \frac{1}{2}) \hbar \omega$ 基态: $E_0 = \frac{1}{2} \hbar \omega$ 激发态: $E_1 = \frac{3}{2} \hbar \omega$

$$\begin{aligned} \psi(x, t) &= e^{-iEt/\hbar} \psi(x, t=0) = e^{-iEt/\hbar} A(2\varphi_0 + \varphi_1) \\ &= \frac{1}{\sqrt{5}} (2e^{-i\omega t/2} \varphi_0 + e^{-3i\omega t/2} \varphi_1) \end{aligned}$$

(3) 当 $t=0$ 时: $\langle E \rangle = \frac{4}{5} E_0 + \frac{1}{5} E_1 = \frac{4}{5} \cdot \frac{1}{2} \hbar \omega + \frac{1}{5} \cdot \frac{3}{2} \hbar \omega = \frac{7}{10} \hbar \omega$

当 $t \neq 0$ 时: $\psi(x, t)$ 仍是归一化的。 φ_0, φ_1 归一化系数仍为 $\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}$ 。

$$\therefore \langle E \rangle = \frac{7}{10} \hbar \omega$$

(4) $\langle x \rangle = \langle \psi | x | \psi \rangle = \frac{1}{5} \int (2\langle \varphi_0 | + \langle \varphi_1 |) x (2|\varphi_0\rangle + |\varphi_1\rangle) dx$

已知: $x|\varphi_n\rangle = \frac{1}{\alpha} \left[\sqrt{\frac{n}{2}} |\varphi_{n-1}\rangle + \sqrt{\frac{n+1}{2}} |\varphi_{n+1}\rangle \right]$

$$\therefore x|2\varphi_0\rangle = \frac{2}{\alpha} \sqrt{\frac{1}{2}} |\varphi_1\rangle = \frac{\sqrt{2}}{\alpha} |\varphi_1\rangle \quad x|\varphi_1\rangle = \frac{1}{\alpha} \left(\sqrt{\frac{1}{2}} |\varphi_0\rangle + \sqrt{\frac{2}{2}} |\varphi_2\rangle \right)$$

$$\begin{aligned} \therefore \langle x \rangle &= \frac{1}{5} \int (2\langle \varphi_0 | + \langle \varphi_1 |) \left(\frac{\sqrt{2}}{\alpha} |\varphi_1\rangle + \frac{1}{\sqrt{2}\alpha} |\varphi_0\rangle + \frac{1}{\alpha} |\varphi_2\rangle \right) dx \\ &= \frac{1}{5} \int \frac{\sqrt{2}}{\alpha} |\varphi_0|^2 + \frac{\sqrt{2}}{\alpha} |\varphi_1|^2 dx = \frac{2\sqrt{2}}{5\alpha} \end{aligned}$$

1.24

二、质量为 μ 的粒子处于 $0 \leq x \leq a$ 的一维无限深势阱中，在初始时刻的波函数为

$$\psi(x, t=0) = A \left(1 + \cos \frac{\pi x}{a} \right) \sin \frac{\pi x}{a}$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2\mu a^2} \quad n=1, 2, \dots$$

(1) 求归一化系数 A 。

(2) 求 $t=T$ 时刻，体系的能量。

(3) 求 $t>0$ 时刻的波函数 $\psi(x, t)$ 。

(4) 求在 t_0 时刻，粒子处于 $0 \leq x \leq \frac{a}{2}$ 的概率。

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$$

解: (1) 归一化 $|\psi|^2 = 1$

$$\begin{aligned} \int \psi^* \psi dx &= \int A^2 \left[\left(1 + \cos \frac{\pi x}{a} \right) \sin \frac{\pi x}{a} \right] \left(1 + \cos \frac{\pi x}{a} \right) \sin \frac{\pi x}{a} dx \\ &= |A|^2 \int \left(\frac{5}{8} - \frac{1}{2} \cos \frac{2\pi x}{a} - \frac{1}{8} \cos \frac{4\pi x}{a} \right) dx \\ &= \frac{5}{8} a |A|^2 = 1 \end{aligned}$$

$$\int \cos \frac{\pi x}{a} \sin^2 \frac{\pi x}{a} dx = 0$$

$$\therefore A = \sqrt{\frac{8}{5a}}$$

$$(2) \psi(x, t=0) = \sqrt{\frac{8}{5a}} \sin \frac{\pi x}{a} + \sqrt{\frac{8}{5a}} \cos \frac{\pi x}{a} \sin \frac{\pi x}{a}$$

$$= \sqrt{\frac{4}{5}} \sqrt{\frac{2}{a}} \sin \frac{\pi x}{a} + \sqrt{\frac{1}{5}} \sqrt{\frac{2}{a}} \sin \frac{2\pi x}{a} = \sqrt{\frac{4}{5}} \psi_1(x) + \sqrt{\frac{1}{5}} \psi_2(x)$$

$$\therefore \langle E \rangle = \frac{4}{5} E_1 + \frac{1}{5} E_2 = \frac{4}{5} \cdot \frac{\pi^2 \hbar^2}{2\mu a^2} + \frac{1}{5} \cdot \frac{4\pi^2 \hbar^2}{2\mu a^2} = \frac{4\pi^2 \hbar^2}{5\mu a^2}$$

在 $t=T$ 时，体系能量与初始时刻能量一样，为 $\frac{4\pi^2 \hbar^2}{5\mu a^2}$

$$\begin{aligned} (3) \psi(x, t) &= \sqrt{\frac{4}{5}} e^{-iE_1 t/\hbar} \psi_1(x) + \sqrt{\frac{1}{5}} e^{-iE_2 t/\hbar} \psi_2(x) \\ &= \sqrt{\frac{4}{5}} e^{-iE_1 t/\hbar} \psi_1(x) + \sqrt{\frac{1}{5}} e^{-iE_2 t/\hbar} \psi_2(x) \end{aligned}$$

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} \quad E_n = \frac{n^2 \pi^2 \hbar^2}{2\mu a^2}$$

$$(4) \int_0^{\frac{a}{2}} |\psi(x, t_0)|^2 dx = \int_0^{\frac{a}{2}} \left[\frac{4}{5} |\psi_1(x)|^2 + \frac{1}{5} |\psi_2(x)|^2 + \frac{4}{5a} \sin \frac{\pi x}{a} \sin \frac{2\pi x}{a} \cdot \cos \frac{(E_2 - E_1)t}{\hbar} \right] dx$$

$$\begin{aligned} &= \int_0^{\frac{a}{2}} \left[\frac{4}{5} \psi_1^2(x) + \frac{1}{5} \psi_2^2(x) + \frac{4}{5a} \cos \frac{3\pi^2 \hbar t}{2\mu a^2} \left(\cos \frac{\pi x}{a} - \cos \frac{3\pi x}{a} \right) \right] dx \\ &= \frac{1}{2} + \frac{16}{15} \cos \frac{3\pi^2 \hbar t}{2\mu a^2} \end{aligned}$$

三、有一质量为 m 的粒子在长度为 L 的圆环上自由运动。

(1) 求定态波函数和能级。 平面转子问题。

(2) 在 $t=0$ 时刻，体系的初始波函数为 $\psi(\varphi, t=0) = A \sin^2 \varphi$ ，求 $t>0$ 时刻的波函数 $\psi(\varphi, t)$ 。

(3) 若体系受到微扰 $H' = \lambda \sin \varphi \cos \varphi$ 的作用，其中 λ 为常数，计算能级的一级修正。
「分类讨论简并和非简并微扰的问题（如果回忆是正确的）」

解：(1) $\hat{H} = \frac{\hat{L}_z^2}{2I} = -\frac{\hbar^2}{2I} \frac{d^2}{d\varphi^2}$

$\hat{L}_z = -i\hbar \frac{d}{d\varphi}$ $\hat{L}_z \psi = l_z \psi$, $-i\hbar \frac{d\psi}{d\varphi} = l_z \psi$

$\int \frac{d\psi}{\psi} = \int \frac{i l_z}{\hbar} d\varphi \quad \therefore \ln \psi = \frac{i l_z}{\hbar} \varphi \quad \therefore \psi = e^{i l_z \varphi / \hbar}$

$\therefore$ 初始条件 $\psi(0) = \psi(2\pi)$ 圆环。

$e^{i l_z 2\pi / \hbar} = \cos \frac{l_z 2\pi}{\hbar} + i \sin \frac{l_z 2\pi}{\hbar} = 1$

$\therefore \cos \frac{l_z 2\pi}{\hbar} = 1 \quad \sin \frac{l_z 2\pi}{\hbar} = 0$

$\therefore l_z = m\hbar, \quad m=0, \pm 1, \pm 2, \dots$

$\therefore \psi = C e^{i m \varphi}$ 归一化得： $C = \frac{1}{\sqrt{2\pi}}$

$\therefore \psi = \frac{1}{\sqrt{2\pi}} e^{i m \varphi}, \quad E = \frac{m^2 \hbar^2}{2I} \quad m=0, \pm 1, \pm 2, \dots$

(2) 归一化： $\int_0^{2\pi} |A|^2 \sin^4 \varphi d\varphi = \frac{3\pi}{4} |A|^2 = 1$

$\therefore A = \sqrt{\frac{4}{3\pi}}$

$\therefore \psi(\varphi, t=0) = A \sin^2 \varphi = A \left[\frac{1}{2i} (e^{i\varphi} - e^{-i\varphi}) \right]^2 = \frac{1}{4} A (e^{2i\varphi} - e^{-2i\varphi} + 2)$

$= \frac{\sqrt{4}}{\sqrt{3\pi}} \frac{\frac{1}{4} \sqrt{\frac{1}{2\pi}} (e^{2i\varphi} + 2e^{0i\varphi} - e^{-2i\varphi})}{\sqrt{\frac{1}{2\pi}}} = \frac{1}{\sqrt{6}} (-\frac{1}{2} + 2\frac{1}{2} - \frac{1}{2})$

$$\begin{aligned}
 \text{当 } t > 0 \text{ 时 } \psi(\psi, t > 0) &= \frac{1}{\sqrt{6}} (-e^{-i\tilde{H}t/\hbar} \psi_2 + 2e^{-i\tilde{H}t/\hbar} \psi_0 - e^{-i\tilde{H}t/\hbar} \psi_2) \\
 &= \frac{1}{\sqrt{6}} (-e^{iE_0 t/\hbar} |\psi_2\rangle + 2e^{-iE_0 t/\hbar} |\psi_0\rangle - e^{-iE_2 t/\hbar} |\psi_2\rangle)
 \end{aligned}$$

其中 $E_2 = \frac{4\hbar^2}{2\mu L^2}$ $E_0 = 0$ $E_2 = \frac{4\hbar^2}{2\mu L^2}$.

13) $H' = \lambda \sin y \cos y$ 求级能量修正.

当 $m=0$ 时. $\psi_0(\psi) = \frac{1}{\sqrt{2\pi}}$ 能量为非简并态.

$$\begin{aligned}
 E_0^{(1)} &= \langle \psi_0 | H' | \psi_0 \rangle = \int \frac{1}{2\pi} \lambda \sin y \cos y dy = \frac{\lambda}{2\pi} \int \frac{\Delta}{2} \sin 2y dy \\
 &= \frac{\lambda}{4\pi} \int_0^{2\pi} \sin 2y dy \\
 &= \frac{\lambda}{4\pi} \left(-\frac{1}{2} \cos 2y \right) \Big|_0^{2\pi} \\
 &= 0.
 \end{aligned}$$

当 $m \neq 0$ 时.

$$\psi_1 = \psi_m^{(0)}(\psi) = \frac{1}{\sqrt{2\pi}} e^{im\psi} \quad \psi_2 = \psi_m^{(0)}(\psi) = \frac{1}{\sqrt{2\pi}} e^{-im\psi}$$

在 E 的本征空间中给出 H' 的矩阵

$$H'_{11} = \langle \psi_1 | H' | \psi_1 \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{-im\psi} \frac{\lambda}{2} \sin 2\psi e^{im\psi} dy = 0.$$

$$H'_{22} = \langle \psi_2 | H' | \psi_2 \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{+im\psi} \frac{\lambda}{2} \sin 2\psi e^{-im\psi} dy = 0.$$

$$\begin{aligned}
 H'_{12} &= \langle \psi_1 | H' | \psi_2 \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{-im\psi} \frac{\lambda}{2} \sin 2\psi e^{-im\psi} dy = \\
 &= \frac{\lambda}{4\pi} \int_0^{2\pi} \sin 2\psi e^{-i2m\psi} dy \\
 &= \frac{\lambda}{4\pi} \int_0^{2\pi} \sin 2\psi (\cos 2m\psi - i \sin 2m\psi) dy \\
 &= \frac{\lambda}{4\pi} \left[\frac{1}{2} \int_0^{2\pi} (\sin(2+2m)\psi + \sin(2-2m)\psi) dy + \right. \\
 &\quad \left. \frac{i}{2} (\cos(2-2m)\psi - \cos(2+2m)\psi) dy \right]
 \end{aligned}$$

$$\therefore H'_{21} = H'_{12}^* = 0$$

$\therefore m \neq 0$ 时 一级修正为 0. $\therefore$ 综上, 能量一级修正为 0.

四、原处于基态的氢原子受到一在方向一致空间均匀，但随时间衰减的电场的影响

5.47

$$\varepsilon(t) = \begin{cases} 0, & t < 0 \\ \varepsilon_0 e^{-t/\tau}, & t > 0 \end{cases}$$

其中 ε_0 , $\tau > 0$ 为常数。取电场方向为 z 轴方向，电场对氢原子的影响视为微扰

$H' = -q\vec{r} \cdot \vec{E}$ 已知

引入电场所带来的电势能

电子 $q = -e$

$$\psi_{100} = \sqrt{\frac{1}{\pi a^3}} e^{-r/a}, \quad \psi_{210} = \frac{1}{4a^2} \sqrt{\frac{1}{2\pi a}} r e^{-r/2a} \cos \theta$$

$$\psi_{21\pm 1} = \mp \frac{1}{8a^2} \sqrt{\frac{1}{\pi a}} r e^{-r/2a} \sin \theta e^{\pm i\phi}$$

(1) 求 $\langle 21m | H' | 100 \rangle$ 的矩阵元。

(2) 当 $t \gg \tau$ 时，求基态氢原子跃迁到 $2p$ 的概率。

解: (1) 微扰 $H' = -q\vec{r} \cdot \vec{E} = e\varepsilon_0 e^{-t/\tau} z = e\varepsilon_0 e^{-t/\tau} r \cos \theta$

$$\begin{aligned} \text{当 } m = \pm 1 \text{ 时 } \langle 21m | H' | 100 \rangle &= \int \psi_{21\pm 1}^* H' \psi_{100} d\tau \\ &= \mp \frac{1}{8a^2} \sqrt{\frac{1}{2\pi a}} \sqrt{\frac{1}{\pi a^3}} \iiint r e^{-r/2a} \sin \theta e^{\mp i\phi} e\varepsilon_0 e^{-t/\tau} r \cos \theta e^{-r/a} r^2 \sin \theta dr d\theta d\phi \end{aligned}$$

$$\text{其中 } I_\phi = \int_0^{2\pi} e^{\mp i\phi} d\phi = 0$$

$$\therefore \langle 21\pm 1 | H' | 100 \rangle = 0$$

$$\text{当 } m = 0 \text{ 时 } \langle 210 | H' | 100 \rangle = \int \psi_{210}^* H' \psi_{100} d\tau$$

$$\begin{aligned} &= \frac{1}{4a^2} \sqrt{\frac{1}{2\pi a}} \sqrt{\frac{1}{\pi a^3}} \iiint r e^{-r/2a} \cos \theta \cdot e\varepsilon_0 e^{-t/\tau} r \cos \theta \cdot e^{-r/a} r^2 \sin \theta dr d\theta d\phi \\ &= \frac{e\varepsilon_0 e^{-t/\tau}}{4\pi\sqrt{2} a^4} \int_0^\infty e^{-3r/2a} r^4 dr \int_0^\pi \cos^2 \theta \sin \theta d\theta \int_0^{2\pi} d\phi \end{aligned}$$

$$I_r = \int_0^\infty e^{-\frac{3r}{2a}} r^4 dr = \frac{4!}{(\frac{3}{2a})^5} = \frac{2^8 a^5}{3^4}$$

$$\begin{aligned} I_\theta &= -\int_0^\pi \cos^3 \theta d\cos \theta \\ &= -\frac{1}{3} \cos^3 \theta \Big|_0^\pi = \frac{2}{3} \end{aligned}$$

$$\therefore \langle 210 | H' | 100 \rangle = \frac{2^7 \sqrt{2} e\varepsilon_0 e^{-t/\tau}}{3^5} a$$

(2) 跃迁公式

$$W_{(100) \rightarrow (21m)}(t) = \frac{1}{\hbar^2} \left| \int_0^t H'_{(21m), (100)}(t) e^{i\omega_{21}} dt \right|^2$$

$$\omega_{21} = \frac{E_2 - E_1}{\hbar} \quad H'_{(211), (100)} = 0 \quad H'_{(210), (100)} = \frac{2^7 \sqrt{2} e \epsilon_0 e^{-t/\tau} a}{3^5}$$

$$\therefore W_{(100) \rightarrow (211)} = 0.$$

$$\begin{aligned} W_{(100) \rightarrow (210)} &= \frac{1}{\hbar^2} \left| \int_0^t \frac{2^7 \sqrt{2} e \epsilon_0 e^{-t/\tau} a}{3^5} e^{i\omega_{21} t} dt \right|^2 \\ &= \frac{2^{15} e^2 \epsilon_0^2 a^2}{3^{10} \hbar^2} \left| \int_0^t e^{(i\omega_{21} - \frac{1}{\tau})t} dt \right|^2 \\ &= \frac{2^{15} e^2 \epsilon_0^2 a^2}{3^{10} \hbar^2} \frac{1}{(\frac{1}{\tau})^2 + (\omega_{21})^2} \\ &= \frac{2^{15} e^2 \epsilon_0^2 a^2 \tau^2}{3^{10} \hbar^2 [1 + (\frac{3e^2 \tau}{8a\hbar})^2]} \end{aligned}$$

$$\begin{aligned} \int_0^t e^{(i\omega_{21} - \frac{1}{\tau})t} dt &= \frac{1}{i\omega_{21} - \frac{1}{\tau}} e^{(i\omega_{21} - \frac{1}{\tau})t} \Big|_0^t \\ &= \frac{e^{i\omega_{21} t} e^{-t/\tau}}{i\omega_{21} - \frac{1}{\tau}} - \frac{1}{i\omega_{21} - \frac{1}{\tau}} \\ &= \frac{e^{i\omega_{21} t} e^{-t/\tau} - 1}{i\omega_{21} - \frac{1}{\tau}} \end{aligned}$$

$$\because t \gg \tau, \quad e^{-t/\tau} \rightarrow e^{-\infty} \rightarrow 0.$$

$$\therefore \left| \int_0^t e^{(i\omega_{21} - \frac{1}{\tau})t} dt \right|^2 = \left| \frac{-1}{i\omega_{21} - \frac{1}{\tau}} \right|^2 = \frac{1}{(\frac{1}{\tau})^2 + (\omega_{21})^2}$$

五、在均匀磁场中，两电荷异号自旋为 $\frac{1}{2}$ 的粒子的哈密顿量为 $H = H_I + H_B$ ，其中

6.3

$$H_I = -a \hat{S}_1 \hat{S}_2, \quad H_B = -\frac{b}{4\hbar} \hat{S}_{1z} + \frac{b}{4\hbar} \hat{S}_{2z}. \quad \text{试在 } \hat{S}^2, \hat{S}_z \text{ 表象中}$$

(1) 求 H 的本征态和本征值。

解：(1) $H = H_I + H_B = -a \hat{S}_1 \hat{S}_2 + \frac{b}{4\hbar} (\hat{S}_{2z} - \hat{S}_{1z}) = -\frac{a}{2} (\hat{S}^2 - \frac{3}{2} \hbar^2) + \frac{b}{4\hbar} (\hat{S}_{2z} - \hat{S}_{1z})$

耦合表象： $|\psi_1\rangle = |11\rangle$ $|\psi_2\rangle = |1-1\rangle$ $|\psi_3\rangle = |10\rangle$ $|\psi_4\rangle = |00\rangle$

$$\hat{H}|\psi_1\rangle = \left[-\frac{a}{2} (\hat{S}^2 - \frac{3}{2} \hbar^2) + \frac{b}{4\hbar} (\hat{S}_{2z} - \hat{S}_{1z}) \right] |11\rangle$$

$$= -\frac{a}{2} [S(S+1)\hbar^2 - \frac{3}{2} \hbar^2] |11\rangle + \frac{b}{4\hbar} (\hat{S}_{2z} - \hat{S}_{1z}) |\uparrow\uparrow\rangle$$

$$= -\frac{a\hbar^2}{4} |11\rangle + \frac{b}{4\hbar} \left(\frac{\hbar}{2} - \frac{\hbar}{2} \right) |\uparrow\uparrow\rangle = -\frac{a\hbar^2}{4} |11\rangle = -\frac{a\hbar^2}{4} |\psi_1\rangle$$

$$\hat{H}|\psi_2\rangle = \left[-\frac{a}{2} (\hat{S}^2 - \frac{3}{2} \hbar^2) + \frac{b}{4\hbar} (\hat{S}_{2z} - \hat{S}_{1z}) \right] |1-1\rangle$$

$$= -\frac{a}{2} [S(S+1)\hbar^2 - \frac{3}{2} \hbar^2] |1-1\rangle + \frac{b}{4\hbar} (\hat{S}_{2z} - \hat{S}_{1z}) |\downarrow\downarrow\rangle$$

$$= -\frac{a\hbar^2}{4} |1-1\rangle = -\frac{a\hbar^2}{4} |\psi_2\rangle$$

$$\hat{H}|\psi_3\rangle = \left[-\frac{a}{2} (\hat{S}^2 - \frac{3}{2} \hbar^2) + \frac{b}{4\hbar} (\hat{S}_{2z} - \hat{S}_{1z}) \right] |10\rangle$$

$$= -\frac{a\hbar^2}{4} |10\rangle + \frac{b}{4\hbar} (\hat{S}_{2z} - \hat{S}_{1z}) \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

$$= -\frac{a\hbar^2}{4} |10\rangle + \frac{b}{4\hbar\sqrt{2}} \left[\frac{\hbar}{2} - \left(-\frac{\hbar}{2}\right) \right] \frac{1}{\sqrt{2}} |\uparrow\downarrow\rangle + \frac{b}{4\hbar\sqrt{2}} \left(-\frac{\hbar}{2} - \frac{\hbar}{2} \right) \frac{1}{\sqrt{2}} |\downarrow\uparrow\rangle$$

$$= -\frac{a\hbar^2}{4} |10\rangle + \frac{b}{4\hbar\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

$$= -\frac{a\hbar^2}{4} |10\rangle + \frac{b}{4} |00\rangle$$

$$\hat{H}|\psi_4\rangle = \left[-\frac{a}{2} (\hat{S}^2 - \frac{3}{2} \hbar^2) + \frac{b}{4\hbar} (\hat{S}_{2z} - \hat{S}_{1z}) \right] |00\rangle$$

$$= \frac{3a\hbar^2}{4} |00\rangle + \frac{b}{4\hbar} (\hat{S}_{2z} - \hat{S}_{1z}) \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

$$= \frac{3a\hbar^2}{4} |00\rangle + \frac{b}{4\hbar\sqrt{2}} \left[\left(\frac{\hbar}{2} - \left(-\frac{\hbar}{2}\right) \right) \frac{1}{\sqrt{2}} |\uparrow\downarrow\rangle - \left(-\frac{\hbar}{2} - \frac{\hbar}{2} \right) \frac{1}{\sqrt{2}} |\downarrow\uparrow\rangle \right] = \frac{3a\hbar^2}{4} |00\rangle + \frac{b}{4} |10\rangle$$

∴ $|\psi_1\rangle$ 和 $|\psi_2\rangle$ 是 $\hat{H}$ 的本征值为 $-\frac{a\hbar^2}{4}$ 的本征态,
 另外两个本征态应该由 $|\psi_3\rangle$ 和 $|\psi_4\rangle$ 的线性组合而成.

$$\hat{H}\psi = E\psi \quad \psi = C_1|\psi_3\rangle + C_2|\psi_4\rangle$$

$$\text{定态方程} = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$\text{解久期方程} \quad \begin{vmatrix} H_{11} - E & H_{12} \\ H_{21} & H_{22} - E \end{vmatrix} = 0.$$

$$\begin{aligned} \therefore H_{11} &= \langle \psi_3 | H | \psi_3 \rangle = \langle 10 | -\frac{a}{2}(S^2 - \frac{3}{2}\hbar^2) + \frac{b}{4\hbar}(\hat{S}_{2z} - \hat{S}_z) | 10 \rangle \\ &= \langle 10 | (-\frac{a\hbar^2}{4} | 10 \rangle + \frac{b}{4} | 00 \rangle) \\ &= -\frac{a\hbar^2}{4} \end{aligned}$$

$$H_{12} = \langle \psi_3 | H | \psi_4 \rangle = \langle 10 | (\frac{3a\hbar^2}{4} | 00 \rangle + \frac{b}{4} | 10 \rangle) = \frac{b}{4}$$

$$H_{21} = H_{12}^* = \frac{b}{4}$$

$$H_{22} = \langle \psi_4 | H | \psi_4 \rangle = \langle 00 | (\frac{3a\hbar^2}{4} | 00 \rangle + \frac{b}{4} | 10 \rangle) = \frac{3a\hbar^2}{4}$$

$$\therefore \begin{vmatrix} -\frac{a\hbar^2}{4} - E & \frac{b}{4} \\ \frac{b}{4} & \frac{3a\hbar^2}{4} - E \end{vmatrix} = 0 \quad \therefore E^2 - \frac{a\hbar^2}{2}E - (\frac{3a^2\hbar^4}{16} + \frac{b^2}{16}) = 0.$$

$$E = \frac{1}{2} (\frac{1}{2}a\hbar^2 \pm \sqrt{(\frac{a\hbar^2}{2})^2 + \frac{1}{4}(3a^2\hbar^2 + b^2)})$$

$$\frac{1}{2} A^2 = (\frac{a\hbar^2}{2})^2 + \frac{1}{4}(3a^2\hbar^2 + b^2) \quad E = \frac{1}{2} (\frac{1}{2}a\hbar^2 \pm A)$$

$$\therefore E_1 = \frac{1}{2} (\frac{1}{2}a\hbar^2 + A) \quad E_2 = \frac{1}{2} (\frac{1}{2}a\hbar^2 - A)$$

$$\text{分别代入本征方程} \quad -\frac{a\hbar^2}{4}C_1 + \frac{b}{4}C_2 = E_1C_1 \quad C_2 = \frac{E_1 + \frac{a\hbar^2}{4}}{-\frac{b}{4}}C_1$$

$$\psi_1 = -\frac{b}{4}B|\psi_3\rangle + (E_1 + \frac{a\hbar^2}{4})B|\psi_4\rangle \quad B = \frac{1}{\sqrt{(E_1 + \frac{a\hbar^2}{4})^2 + (\frac{b}{4})^2}}$$

$$\text{同理} \quad \psi_2 = (E_2 + \frac{a\hbar^2}{4})D|\psi_3\rangle - \frac{a\hbar^2}{4}D|\psi_4\rangle \quad D = \frac{1}{\sqrt{(\frac{a\hbar^2}{4})^2 + (E_2 + \frac{a\hbar^2}{4})^2}}$$